STACKS, DISCRIMINANTS: PROPOSITION-TATE-MAP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2933-2946 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2948-3163, the complete author proof (214 body lines).

Editorial note. Navigation stubs: 13 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_discriminant.json (pb=9070, stmt 2933-2946, proof 2948-3163).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'proposition-tate-map' also exists in stacks-snapshot-20261011 — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 0FK8: sources/stacks-discriminant/pinned-original-source.tex, source lines 2168–2185. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0BWH: sources/stacks-discriminant/pinned-original-source.tex, source lines 2727–2734. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0BW9: sources/stacks-discriminant/pinned-original-source.tex, source lines 1907–1912. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0FKA: sources/stacks-discriminant/pinned-original-source.tex, source lines 2225–2242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0BWE: sources/stacks-discriminant/pinned-original-source.tex, source lines 2005–2026. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 0FK9: sources/stacks-discriminant/pinned-original-source.tex, source lines 2188–2195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 01ZM: sources/stacks-discriminant/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 094M: sources/stacks-discriminant/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0C3L: sources/stacks-discriminant/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07RU: sources/stacks-discriminant/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 08KQ: sources/stacks-discriminant/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0H9B: sources/stacks-discriminant/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0FKC: sources/stacks-discriminant/pinned-original-source.tex, source lines 2867–2898. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *There exists a unique rule that to every locally quasi-finite syntomic morphism of locally Noetherian schemes $Y \to X$ assigns an isomorphism*

$$c_{Y/X} : \det(NL_{Y/X}) \longrightarrow \omega_{Y/X}$$

satisfying the following two properties

(1) *the section $\delta(NL_{Y/X})$ is mapped to $\tau_{Y/X}$, and*
(2) *the rule is compatible with restriction to opens and with base change.*

Proof. Let us reformulate the statement of the proposition. Consider the category $\mathcal{C}$ whose objects, denoted Y/X, are locally quasi-finite syntomic morphism $Y \to X$ of locally Noetherian schemes and whose morphisms $b/a : Y'/X' \to Y/X$ are commutative diagrams

$$\begin{array}{ccc} Y' & \xrightarrow{b} & Y \\ \downarrow & & \downarrow \\ X' & \xrightarrow{a} & X \end{array}$$

which induce an isomorphism of Y' with an open subscheme of $X' \times_X Y$. The proposition means that for every object Y/X of $\mathcal{C}$ we have an isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$ and for every morphism $b/a : Y'/X' \to Y/X$ of $\mathcal{C}$ we have $b^*c_{Y/X} = c_{Y'/X'}$ via the identifications $b^*\det(NL_{Y/X}) = \det(NL_{Y'/X'})$ and $b^*\omega_{Y/X} = \omega_{Y'/X'}$ described above.

Given Y/X in $\mathcal{C}$ and $y \in Y$ we can find an affine opens $V \subset Y$ and $U \subset X$ with $y \in V$ and $f(V) \subset U$ such that there exists some isomorphism

$$\det(NL_{Y/X})|_V \longrightarrow \omega_{Y/X}|_V$$

mapping $\delta(NL_{Y/X})|_V$ to $\tau_{Y/X}|_V$. This follows from picking affine opens as in Lemma 0BWE part (5), the affine local description of $\delta(NL_{Y/X})$ in Remark 0FKC, and Lemma 0BWH. If the annihilator of the section $\tau_{Y/X}$ is zero, then these local maps are unique and automatically glue. Hence if the annihilator of $\tau_{Y/X}$ is zero, then there is a unique isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$. If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}$ and the annihilator of $\tau_{Y'/X'}$ is zero as well, then $b^*c_{Y/X}$ is the unique isomorphism $c_{Y'/X'} : \det(NL_{Y'/X'}) \to \omega_{Y'/X'}$ with $c_{Y'/X'}(\delta(NL_{Y'/X'})) = \tau_{Y'/X'}$. This follows formally from the fact that $b^*\delta(NL_{Y/X}) = \delta(NL_{Y'/X'})$ and $b^*\tau_{Y/X} = \tau_{Y'/X'}$.

We can summarize the results of the previous paragraph as follows. Let $\mathcal{C}_{nice} \subset \mathcal{C}$ denote the full subcategory of Y/X such that the annihilator of $\tau_{Y/X}$ is zero. Then we have solved the problem on $\mathcal{C}_{nice}$. For Y/X in $\mathcal{C}_{nice}$ we continue to denote $c_{Y/X}$ the solution we've just found.

Consider morphisms

$$Y_1/X_1 \xleftarrow{b_1/a_1} Y/X \xrightarrow{b_2/a_2} Y_2/X_2$$

in $\mathcal{C}$ such that Y_1/X_1 and Y_2/X_2 are objects of $\mathcal{C}_{nice}$. **Claim.** $b_1^*c_{Y_1/X_1} = b_2^*c_{Y_2/X_2}$. We will first show that the claim implies the proposition and then we will prove the claim.

Let $d, n \geq 1$ and consider the locally quasi-finite syntomic morphism $Y_{n,d} \to X_{n,d}$ constructed in Example 0FK8. Then $Y_{n,d}$ is an irreducible regular scheme and the morphism $Y_{n,d} \to X_{n,d}$ is locally quasi-finite syntomic and étale over a dense open, see Lemma 0FK9. Thus $\tau_{Y_{n,d}/X_{n,d}}$ is nonzero for example by Lemma 0BW9. Now a nonzero section of an invertible module over an irreducible regular scheme has vanishing annihilator. Thus $Y_{n,d}/X_{n,d}$ is an object of $\mathcal{C}_{nice}$.

Let Y/X be an arbitrary object of $\mathcal{C}$. Let $y \in Y$. By Lemma 0FKA we can find $n, d \geq 1$ and morphisms

$$Y/X \leftarrow V/U \xrightarrow{b/a} Y_{n,d}/X_{n,d}$$

of $\mathcal{C}$ such that $V \subset Y$ and $U \subset X$ are open. Thus we can pullback the canonical morphism $c_{Y_{n,d}/X_{n,d}}$ constructed above by b to V. The claim guarantees these local isomorphisms glue! Thus we get a well defined global isomorphism $c_{Y/X} : \det(NL_{Y/X}) \to \omega_{Y/X}$ with $c_{Y/X}(\delta(NL_{Y/X})) = \tau_{Y/X}$. If $b/a : Y'/X' \to Y/X$ is a morphism of $\mathcal{C}$, then the claim also implies that the similarly constructed map $c_{Y'/X'}$ is the pullback by b of the locally constructed map $c_{Y/X}$. Thus it remains to prove the claim.

In the rest of the proof we prove the claim. We may pick a point $y \in Y$ and prove the maps agree in an open neighbourhood of y. Thus we may replace Y_1, Y_2 by open neighbourhoods of the image of y in Y_1 and Y_2. Therefore we may assume

Y, X, Y_1, X_1, Y_2, X_2 are affine, say they are the spectra of rings B, A, B_1, A_1, B_2, A_2. Picture

$$\begin{array}{ccccc} B_1 & \longrightarrow & B & \longleftarrow & B_2 \\ \uparrow & & \uparrow & & \uparrow \\ A_1 & \longrightarrow & A & \longleftarrow & A_2 \end{array}$$

By assumption the spectrum of B is an affine open of both the spectrum of $A \otimes_{A_1} B_1$ and $A \otimes_{A_2} B_2$. Shrinking more we may assume there exist elements $g_i \in A \otimes_{A_i} B_i$ such that our maps give isomorphisms $(A \otimes_{A_i} B_i)_{g_i} = B$, see Properties, Lemma 0H9B. Let x_α, y_β be a sufficiently large collection of variables such that we may choose surjections

$$A_1' = A_1[x_\alpha] \to A \quad \text{and} \quad A_2' = A_2[y_\beta] \to A$$

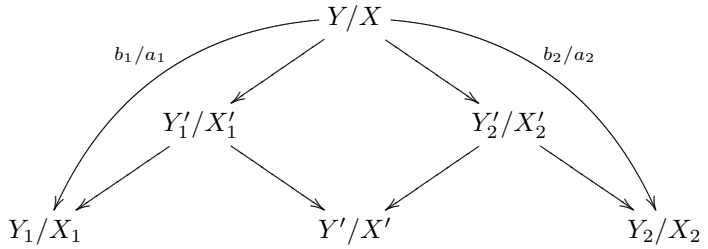
of A_1 and A_2-algebras. Then we can choose lifts $h_i \in A_i' \otimes_{A_i} B_i$ of $g_i \in A \otimes_{A_i} B_i$ and we consider the diagram

$$\begin{array}{ccccc} (A_1' \otimes_{A_1} B_1)_{h_1} & \longrightarrow & B & \longleftarrow & (A_2' \otimes_{A_1} B_2)_{h_2} \\ \uparrow & & \uparrow & & \uparrow \\ A_1' & \longrightarrow & A & \longleftarrow & A_2' \end{array}$$

By construction the two squares are cocartesian. Next, we consider the ring map

$$A' = A_1' \times_A A_2' \longrightarrow B' = (A_1' \otimes_{A_1} B_1)_{h_1} \times_B (A_2' \otimes_{A_1} B_2)_{h_2}$$

By More on Algebra, Lemma 07RU we have $A_1' \otimes_{A'} B' = (A_1' \otimes_{A_1} B_1)_{h_1}$ and $A_2' \otimes_{A'} B' = (A_2' \otimes_{A_2} B_2)_{h_2}$. In particular the fibres of the morphism $Y' = \mathrm{Spec}(B') \to \mathrm{Spec}(A') = X'$ are open subschemes of base changes of the fibres of the maps $Y_i \to X_i$, hence finite and local complete intersections (see Lemma 0BWE). By More on Algebra, Lemma 08KQ the ring map $A' \to B'$ is flat and of finite presentation. Thus by Lemma 0BWE part (3) we conclude that $Y' \to X'$ is locally quasi-finite and syntomic. Thus we obtain the commutative diagram

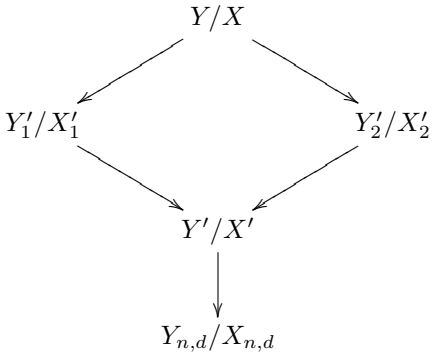


where Y_i'/X_i' corresponds to $A_i' \to (A_i' \otimes_{A_i} B_i)_{h_i}$.

In this paragraph, we fix the issue that A' and A_i' may not be Noetherian by a limit argument; we suggest the reader skip this. We can find a finitely generated $\mathbf{Z}$-subalgebra $A'' \subset A'$ and a quasi-finite and syntomic ring map $A'' \to B''$ such that $B' = A' \otimes_{A''} B''$. This follows from Limits, Lemmas 01ZM, 0C3L, and 094M. Then the ring maps $A'' \to A_1' = A_1[x_\alpha]$ will factor through a finite polynomial subalgebra $A_i[x_1, \ldots, x_n$ and the ring map $A'' \to A_2' = A_2[y_\beta]$ will factor through $A_2[y_1, \ldots, y_m]$. After increasing the sets of variables, we may also assume that $h_1 \in A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1$ and $h_2 \in A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2$. Increasing the sets of variables one more time, we can assume that the maps $B'' \to (A_i' \otimes_{A_i} B_i)_{h_i}$ factor through $(A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1)_{h_1}$ and $(A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2)_{h_2}$. Replacing Y', X', Y_1', X_1', Y_2', X_2' by the spectra of B'', A'', $(A_1[x_1, \ldots, x_n] \otimes_{A_1} B_1)_{h_1}$, $A_1[x_1, \ldots, x_n]$,

$(A_2[y_1, \ldots, y_m] \otimes_{A_2} B_2)_{h_2}$, $A_2[y_1, \ldots, y_m]$, we see that we get a diagram as above where all schemes are Noetherian.

Note that Y_i'/X_i' is an object of $\mathcal{C}_{nice}$ as it obtained by taking an open subscheme of the product of an affine space with Y_i/X_i. In particular, the pullback of c_{Y_i/X_i} via (b_i/a_i) is the same as the pullback of $c_{Y_i'/X_i'}$ via $Y/X \to Y_i'/X_i'$. Now we would be done if Y'/X' is an object of $\mathcal{C}_{nice}$ (but this is likely not the case). Namely, then pulling back $c_{Y'/X'}$ around the two sides of the square, we would obtain the desired conclusion. To get around the problem that Y'/X' is not in $\mathcal{C}_{nice}$ we note the arguments above show that, after possibly shrinking all of the schemes $X, Y, X_1', Y_1', X_2', Y_2', X', Y'$ we can find some $n, d \geq 1$, and extend the diagram like so:



and then we can use the already given argument by pulling back from $c_{Y_{n,d}/X_{n,d}}$. This finishes the proof. $\square$